TIMING

21M.370

Digital Instrument Design

Ian Hattwick

REFERENCE

All references in Canvas unless a link is provided

Chris Wilson. “A Tale of Two Clocks”.

https://web.dev/articles/audio-scheduling

Anne Danielson. “Here, There, and Everywhere: three accounts of pulse in D’Angelo’s ‘Left and Right’”

SCHEDULE

Nobby perfomances Apr 1, 6, 8

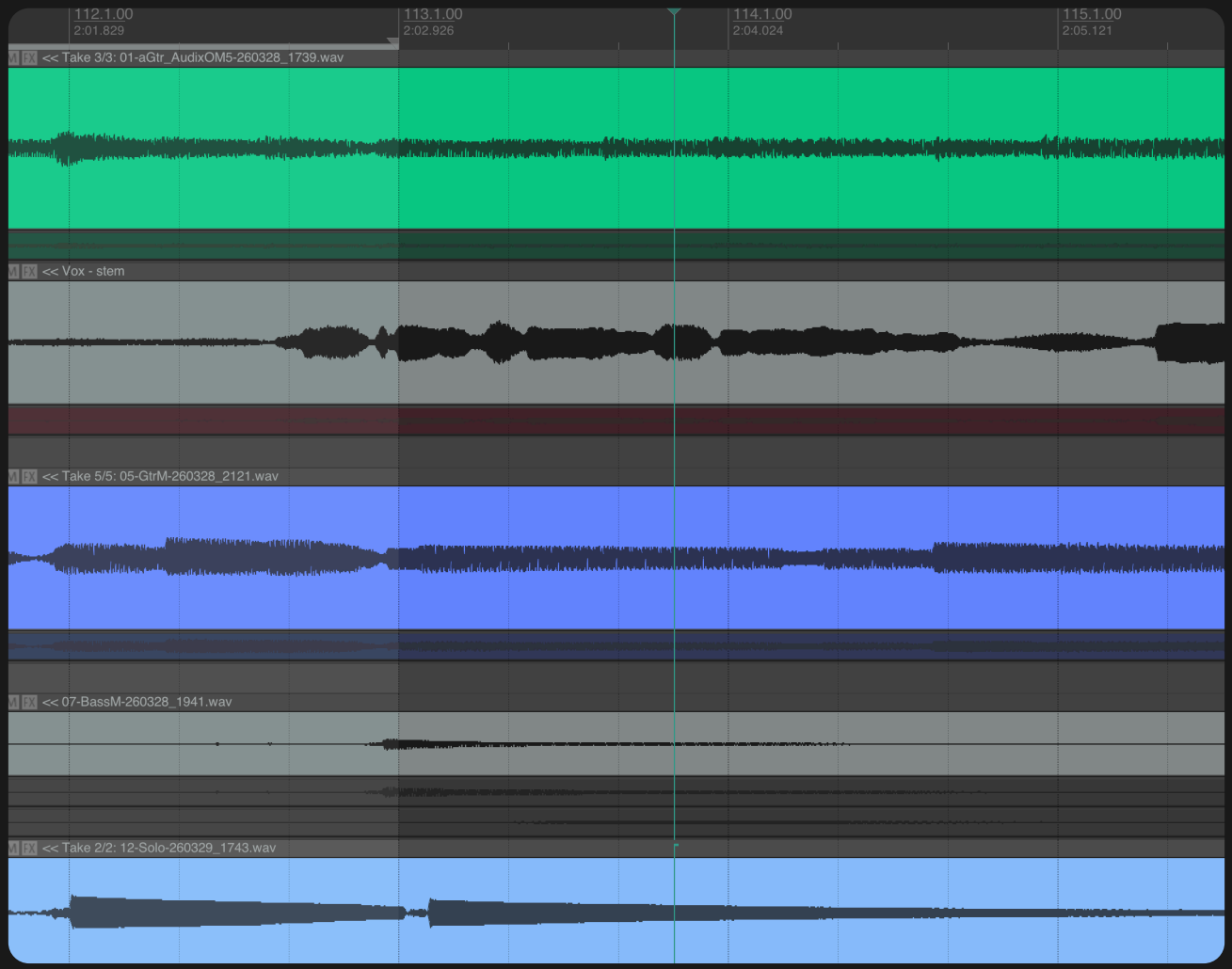
Capacit performances beginning Apr 8

- Andi Otto class visit Apr 8

Performer presentations beginning Apr 6

KINDS OF TIME

Musical time



KINDS OF TIME

Musical time

- generally beat-based
- includes larger groupings (bars, phrases)
- can be tight or loose
- relationships between events is important!
- groove and feel depend on timing, tone, phrasing, etc.

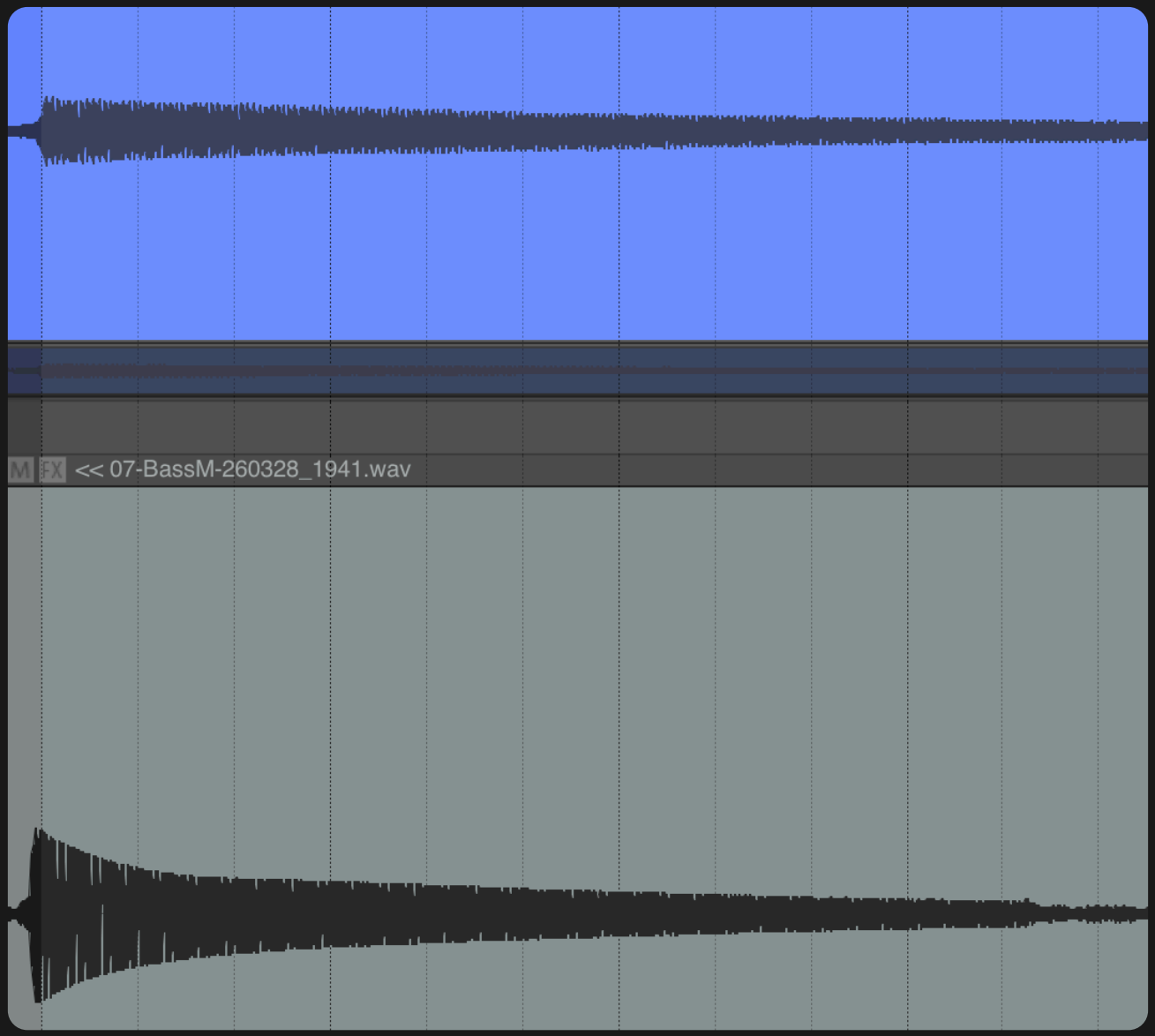
KINDS OF TIME

Real time

- all instruments have time responses, often experienced as the ‘feel’ of an instrument
- piano: travel of key and hammer action
- french horn: 4m of tubing

Physical speed of sound: ~343m/s

- french horn: 4/343 = 0.011s
- sound travel through the air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<< 07-BassM-260328_1941.wav

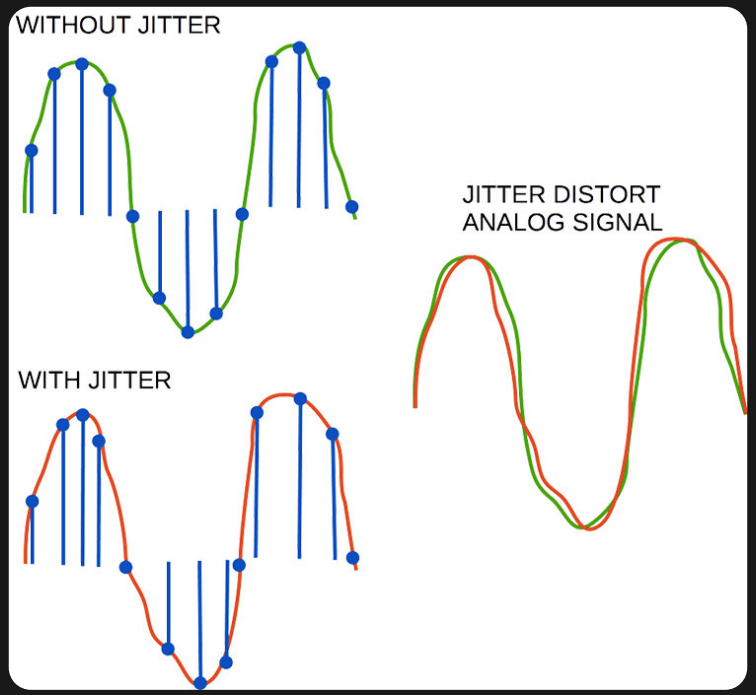
KINDS OF TIME

Computer time

- discrete time
- processes audio in samples, e.g. 48kHz
- requires precision (no jitter) and NO DROPOUTS

In a computer there are two kinds of clocks:

- general purpose clocks: unpredictable timing
- audio clocks (sample-accurate): guaranteed timing



KINDS OF TIME

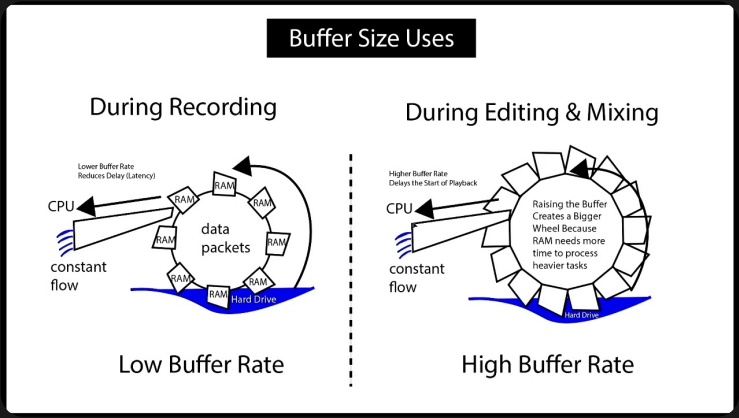
Audio Buffers

- provide protection against dropouts
- minimize CPU cost
- collection of audio samples, powers of 2

Types of buffers

- analog-digital converter (ADC)
- internal software DSP buffer
- digital-analog converter (DAC)

Buffers sacrifice latency for stability



KINDS OF TIME

Communication

- often times not optimized for real-time interaction
- UART / USB 2.0: sends data at bits/second (BAUD rate)
- microcontrollers generally use a UART-USB chip which has its own buffers / latency

Generally, wired and local network comms are OK

Wireless often adds latency UNLESS specifically designed for music, e.g. MIDI BLE

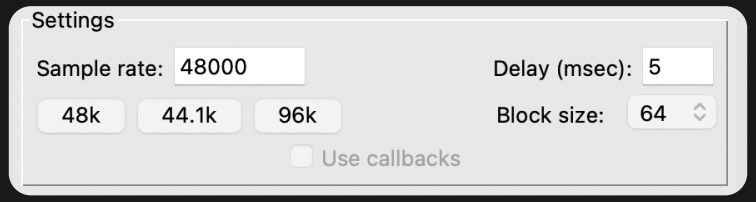
REAL-TIME PERFORMANCE

Latency is how quickly sound is generated in response to action

- ~10ms is considered acceptable . . .
- low jitter is also important

Components:

- ESP32: execution speed, UART communication
- CP2104: USB communication chip
- Python: data processing, OSC messages
- PD: DSP processing, Audio output



AUTOMATING TIME

Keeping good time is hard!

- lots of reasons for latency/ jitter
- being a good musician takes practice!

Computers (can be) good at keeping time

- to guarantee timing accuracy, event timing should be tied to audio clock
- typically python,javascript,etc timing isn’t consistent
- SCAMP in python has a good audio clock
- PD has two types of signals: audio and control
- automitonism uses audio clock for triggers

AUTOMATING TIME

Beat clocks:

- generally pulses-per-quarter-note (ppqn)
- analog: 1 ppqn
- MIDI: 24 ppqn
- how to schedule ‘off-the-grid’?

phrase synchronization

- how to make multiple sequencers play together?
- reset on downbeats (every 8,16,32, etc. beats)

AUTOMATING TIME

Integer clock counters

- increment by 1 every pulse
- sequencers use modulo to determine index, e.g. clock%8 for an eight-step sequence

Schedulers

- sequencers schedule an event in the scheduler
- allows for adjustments to scheduled times

ULTRA-PROCESSED INTERFACES

who was the talk for?

- people who but midi controllers to make music

what are the characteristics of ultra-processed instruments

- make music easier, as a tool
- separated from culture
- music is a problem to be solved

how is culture represented in instruments?

- materials, colors
- feel of the instrument - for guitar string feel, weight, neck size
- idiomatic performance
- what kind of music will be played on it
- where the instrument is played / what compositions / what context
- what parts of the performance are visible to collaborators or audience
- the sound of the instrument
- the scale of the gestures

idiomatic performance

- something that is easy / natural / sounds good on an instrument
- https://youtu.be/fBeQ-O-D3ug